In [2]:

```
import pandas as pd

df = pd.read_csv("C:\\Users\\Jagatjyoti\\Jagat_Code\\21-04-2024\\in+vehicle+coup
print(df.head())   # shows first 5 rows
print(df.shape)        # rows, columns
print(df.columns)      # column names
print(df.describe())   # statistics
print(df["passanger"])       # access a column
```

```
        destination  passanger weather  temperature  time  \
0  No Urgent Place      Alone   Sunny           55   2PM
1  No Urgent Place  Friend(s)   Sunny           80  10AM
2  No Urgent Place  Friend(s)   Sunny           80  10AM
3  No Urgent Place  Friend(s)   Sunny           80   2PM
4  No Urgent Place  Friend(s)   Sunny           80   2PM

                  coupon expiration  gender age      maritalStatus  ...  \
0       Restaurant(<20)         1d  Female  21  Unmarried partner  ...
1          Coffee House         2h  Female  21  Unmarried partner  ...
2  Carry out & Take away         2h  Female  21  Unmarried partner  ...
3          Coffee House         2h  Female  21  Unmarried partner  ...
4          Coffee House         1d  Female  21  Unmarried partner  ...

   CoffeeHouse CarryAway RestaurantLessThan20 Restaurant20To50  \
0        never       NaN                  4~8              1~3
1        never       NaN                  4~8              1~3
2        never       NaN                  4~8              1~3
3        never       NaN                  4~8              1~3
4        never       NaN                  4~8              1~3

  toCoupon_GEQ5min toCoupon_GEQ15min toCoupon_GEQ25min direction_same  \
0                1                 0                 0              0
1                1                 0                 0              0
2                1                 1                 0              0
3                1                 1                 0              0
4                1                 1                 0              0

  direction_opp  Y
0             1  1
1             1  0
2             1  1
3             1  0
4             1  0

[5 rows x 26 columns]
(12684, 26)
Index(['destination', 'passanger', 'weather', 'temperature', 'time', 'coupon',
       'expiration', 'gender', 'age', 'maritalStatus', 'has_children',
       'education', 'occupation', 'income', 'car', 'Bar', 'CoffeeHouse',
       'CarryAway', 'RestaurantLessThan20', 'Restaurant20To50',
       'toCoupon_GEQ5min', 'toCoupon_GEQ15min', 'toCoupon_GEQ25min',
       'direction_same', 'direction_opp', 'Y'],
      dtype='object')
        temperature  has_children  toCoupon_GEQ5min  toCoupon_GEQ15min  \
count  12684.000000  12684.000000           12684.0       12684.000000
mean      63.301798      0.414144               1.0           0.561495
std       19.154486      0.492593               0.0           0.496224
min       30.000000      0.000000               1.0           0.000000
25%       55.000000      0.000000               1.0           0.000000
50%       80.000000      0.000000               1.0           1.000000
75%       80.000000      1.000000               1.0           1.000000
max       80.000000      1.000000               1.0           1.000000

       toCoupon_GEQ25min  direction_same  direction_opp             Y
count       12684.000000    12684.000000   12684.000000  12684.000000
mean            0.119126        0.214759       0.785241      0.568433
std             0.323950        0.410671       0.410671      0.495314
min             0.000000        0.000000       0.000000      0.000000
25%             0.000000        0.000000       1.000000      0.000000
```

```
50%             0.000000        0.000000       1.000000      1.000000
75%             0.000000        0.000000       1.000000      1.000000
max             1.000000        1.000000       1.000000      1.000000
0            Alone
1        Friend(s)
2        Friend(s)
3        Friend(s)
4        Friend(s)
           ...
12679      Partner
12680        Alone
12681        Alone
12682        Alone
12683        Alone
Name: passanger, Length: 12684, dtype: object
```

In [4]:

```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
(244, 7)
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='objec
t')
```

In [42]:

```
sns.get_dataset_names()
```

```
Out[42]: ['anagrams',
 'anscombe',
 'attention',
 'brain_networks',
 'car_crashes',
 'diamonds',
 'dots',
 'dowjones',
 'exercise',
 'flights',
 'fmri',
 'geyser',
 'glue',
 'healthexp',
 'iris',
 'mpg',
 'penguins',
 'planets',
 'seaice',
 'taxis',
 'tips',
 'titanic',
 'anagrams',
 'anagrams',
 'anscombe',
 'anscombe',
 'attention',
 'attention',
 'brain_networks',
 'brain_networks',
 'car_crashes',
 'car_crashes',
 'diamonds',
 'diamonds',
 'dots',
 'dots',
 'dowjones',
 'dowjones',
 'exercise',
 'exercise',
 'flights',
 'flights',
 'fmri',
 'fmri',
 'geyser',
 'geyser',
 'glue',
 'glue',
 'healthexp',
 'healthexp',
 'iris',
 'iris',
 'mpg',
 'mpg',
 'penguins',
 'penguins',
 'planets',
 'planets',
 'seaice',
 'seaice',
```

```
 'taxis',
 'taxis',
 'tips',
 'tips',
 'titanic',
 'titanic',
 'anagrams',
 'anscombe',
 'attention',
 'brain_networks',
 'car_crashes',
 'diamonds',
 'dots',
 'dowjones',
 'exercise',
 'flights',
 'fmri',
 'geyser',
 'glue',
 'healthexp',
 'iris',
 'mpg',
 'penguins',
 'planets',
 'seaice',
 'taxis',
 'tips',
 'titanic']
```

In [6]:

```
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)


import pandas as pd
import seaborn as sns

sns.get_dataset_names()
df = pd.read_csv("C:\\Users\\Jagatjyoti\\Jagat_Code\\22-04-2026\\tips_dataset.cs
print(df.head())   # shows first 5 rows
print(df.shape)        # rows, columns
print(df.columns)      # column names

tips = sns.load_dataset("tips")
tips.head()
```

```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
(244, 7)
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='objec
t')
```

Out[6]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

In [7]:

```
tips = sns.load_dataset("tips")
tips.head()
```

Out[7]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

In [9]:

```
titanic = sns.load_dataset("titanic")
titanic.head()
```

Out[9]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

In [10]:

```
tips
```

Out[10]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

In [11]:
```
sns.set_theme(style="darkgrid")
```

In [15]:
```

```

```
  Cell In[15], line 1
    tips.to_csv("C:\\Users\\Jagatjyoti\\Jagat_Code\\22-04-2026\\tips_dataset.cs
v"\\tips_dataset.csv", index=False)

^
SyntaxError: unexpected character after line continuation character
```

In [17]:
```
tips.to_csv("tips_dataset.csv", index=False)
import pandas as pd
```

In [18]:
```
import os
os.getcwd()
```

Out[18]: 'C:\\Users\\Jagatjyoti'

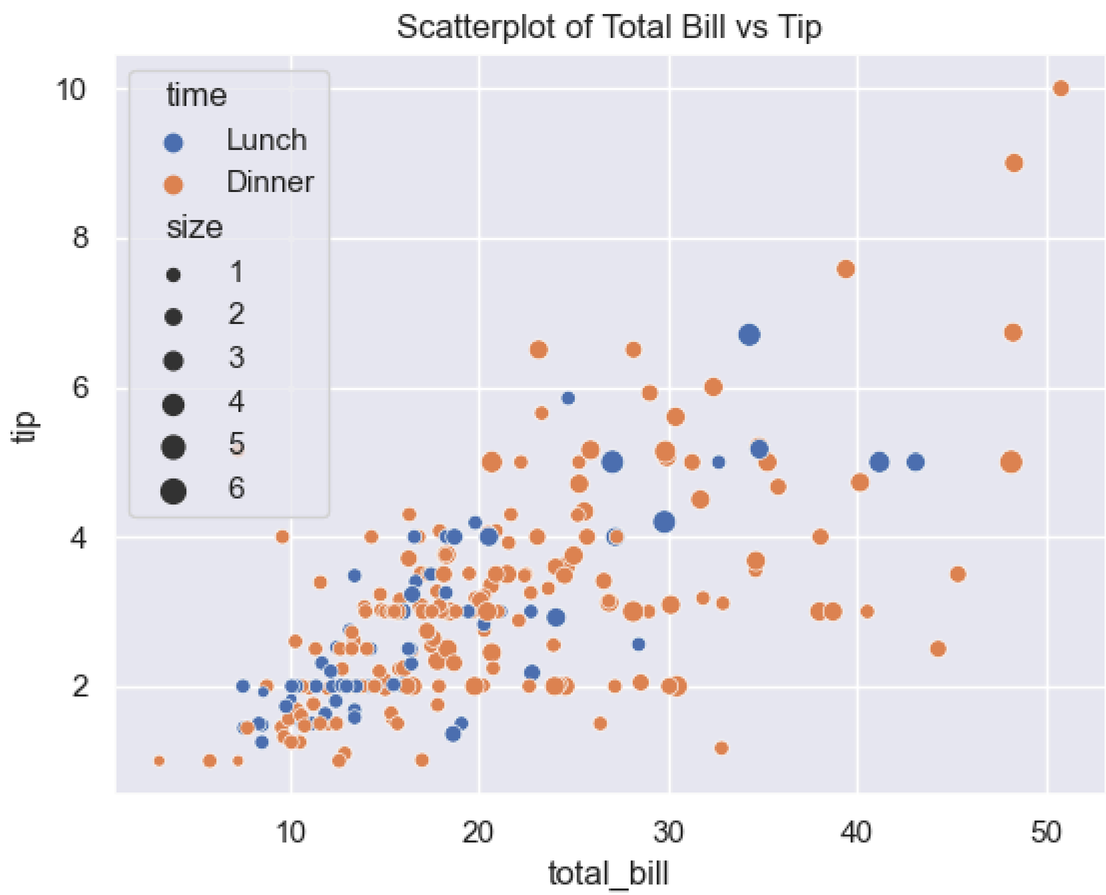
In [19]:
```
import matplotlib.pyplot as plt
```

In [20]:
```
plt.figure(figsize=(8, 6))
```

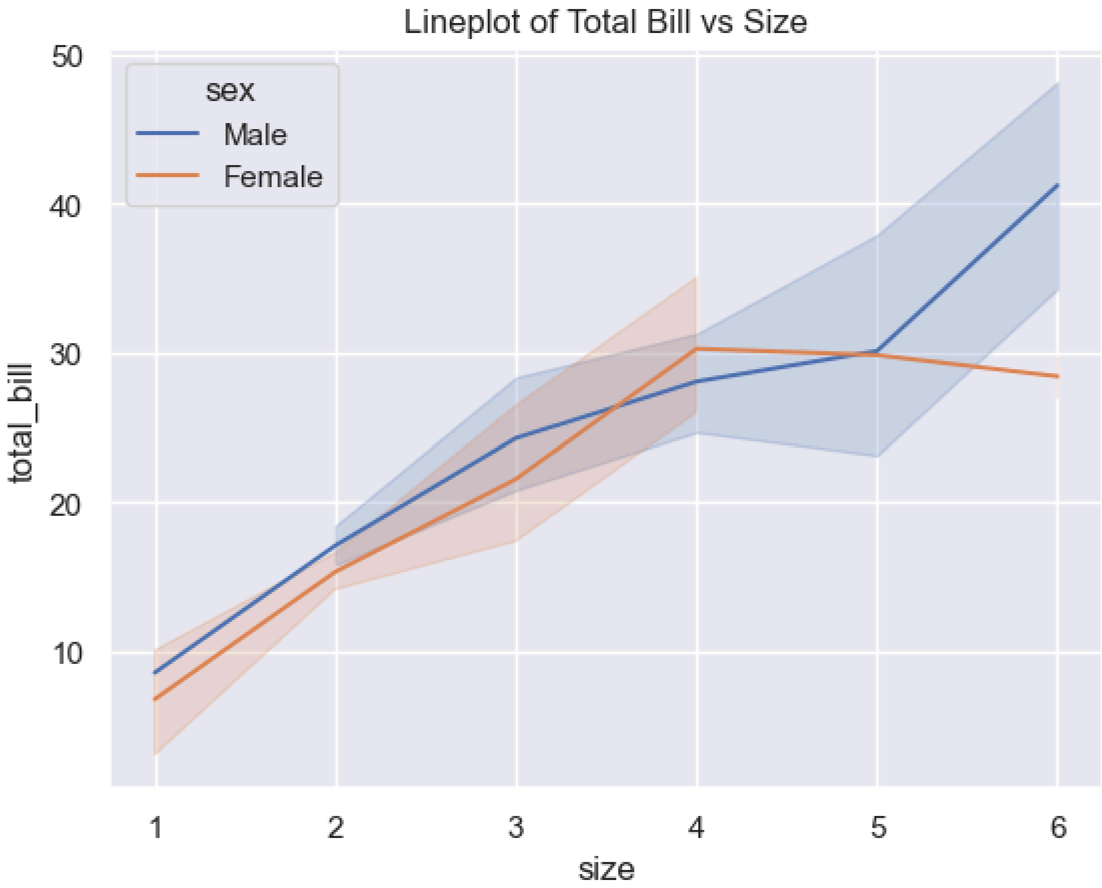
Out[20]: <Figure size 800x600 with 0 Axes>

<Figure size 800x600 with 0 Axes>

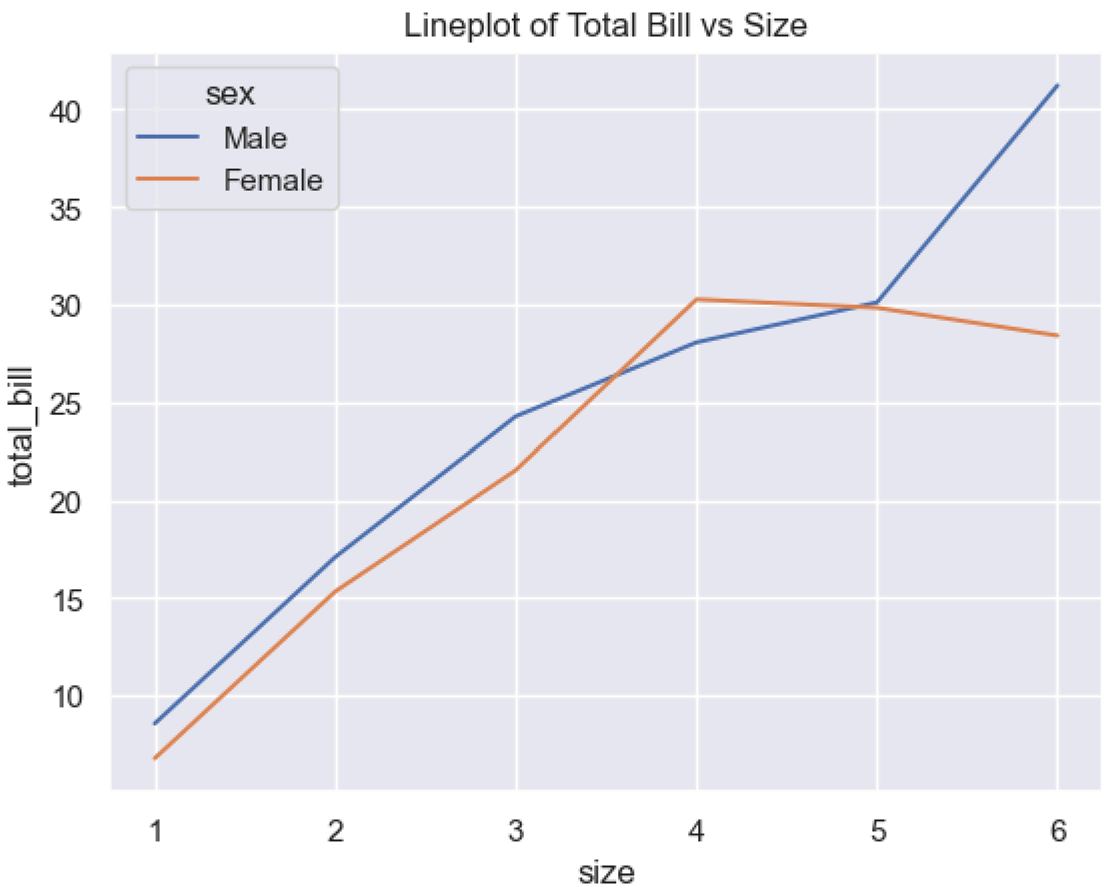
In [21]:
```
# 1. scatter plot

sns.scatterplot(data=tips, x="total_bill", y="tip",hue="time", size="size", pale
plt.title("Scatterplot of Total Bill vs Tip")
plt.show()
```



In [22]:
```
# line plot

sns.lineplot(data=tips, x= 'size', y='total_bill', hue='sex',markers='o')
plt.title("Lineplot of Total Bill vs Size")
plt.show()
```



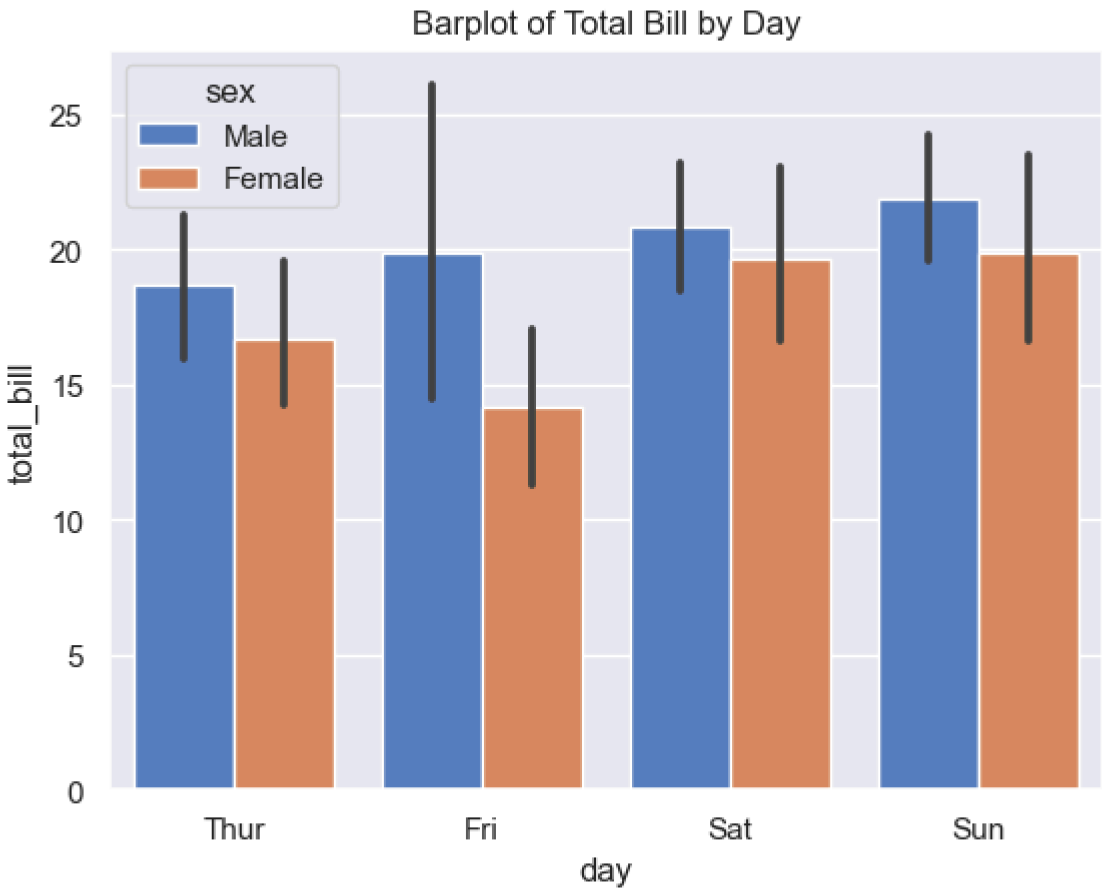
In [23]:
```
sns.lineplot(data=tips, x= 'size', y='total_bill', hue='sex',ci=None, markers='c
plt.title("Lineplot of Total Bill vs Size")
plt.show()
```



In [24]:
```
tips.columns
```

Out[24]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='obj
ect')
```

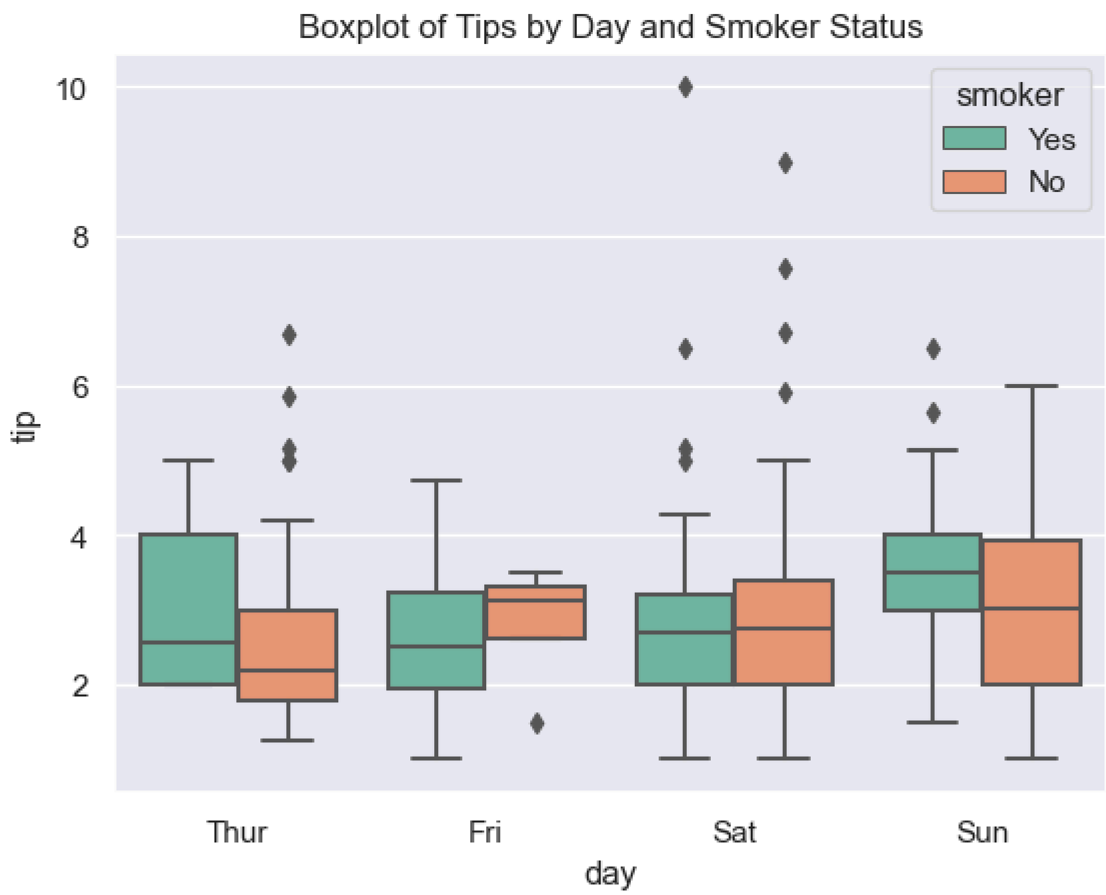
In [25]:
```
#3. bar plot

sns.barplot(data=tips, x='day', y='total_bill', hue = 'sex',palette='muted')
plt.title("Barplot of Total Bill by Day")
plt.show()
```



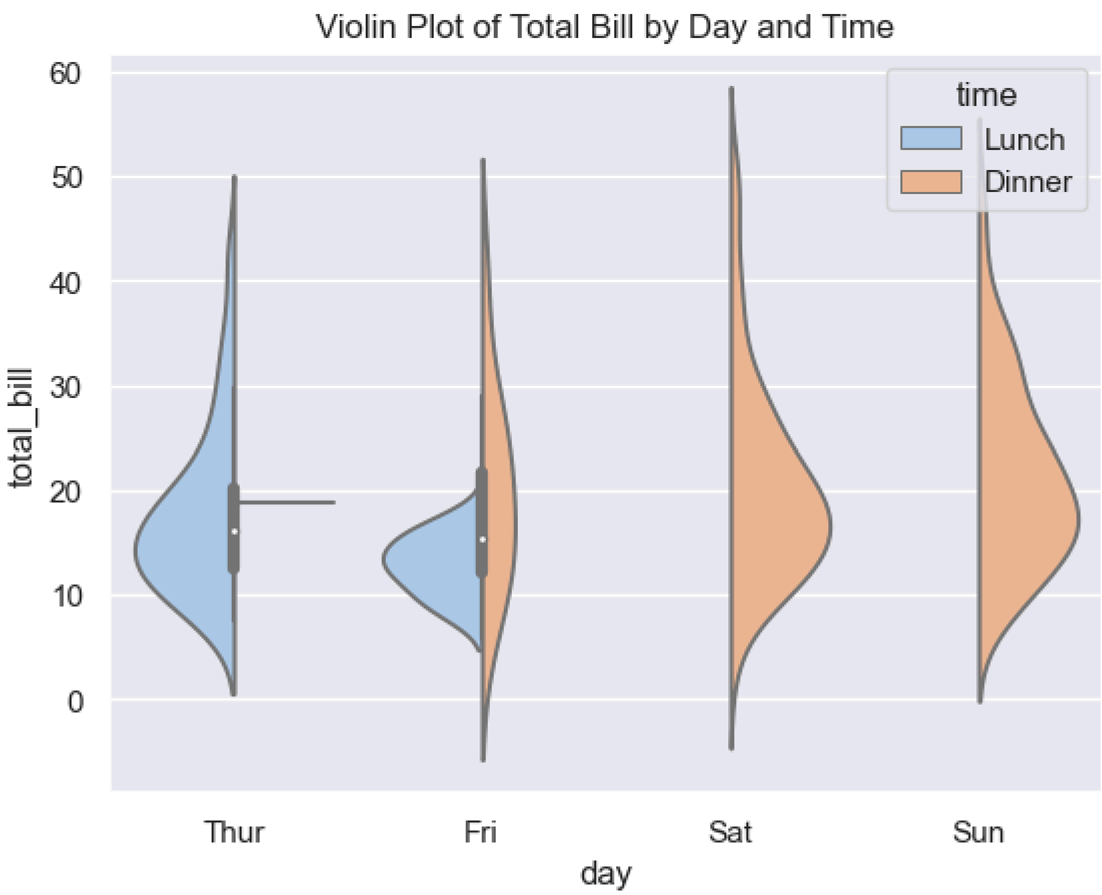
In [26]:
```
tips.columns
```

Out[26]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='obj
ect')
```

In [27]:
```
# 4. Boxplot

sns.boxplot(data=tips, x='day', y='tip', hue='smoker', palette='Set2')
plt.title("Boxplot of Tips by Day and Smoker Status")
plt.show()
```



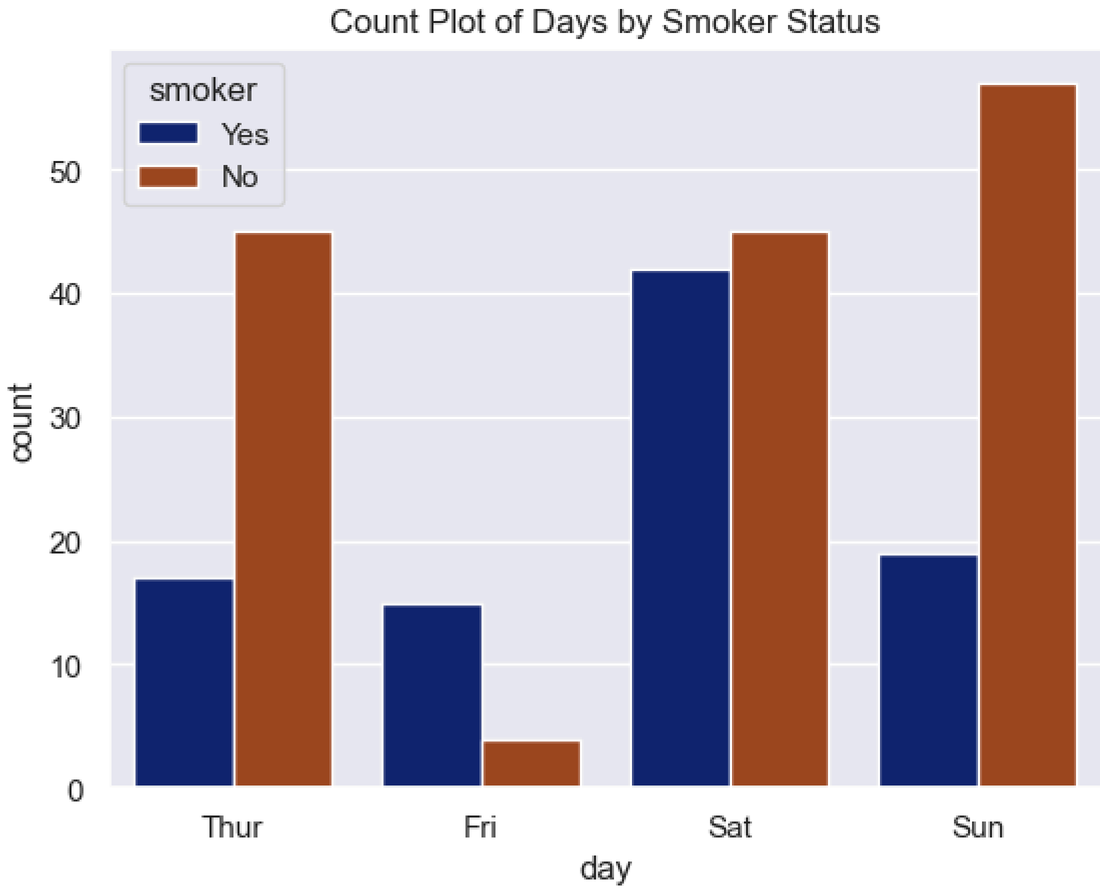
In [28]:
```
# 5. violin plot

sns.violinplot(data=tips, x='day', y='total_bill', hue='time', split=True, palet
plt.title("Violin Plot of Total Bill by Day and Time")
plt.show()
```



In [29]:
```
tips.columns
```

Out[29]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='obj
ect')
```

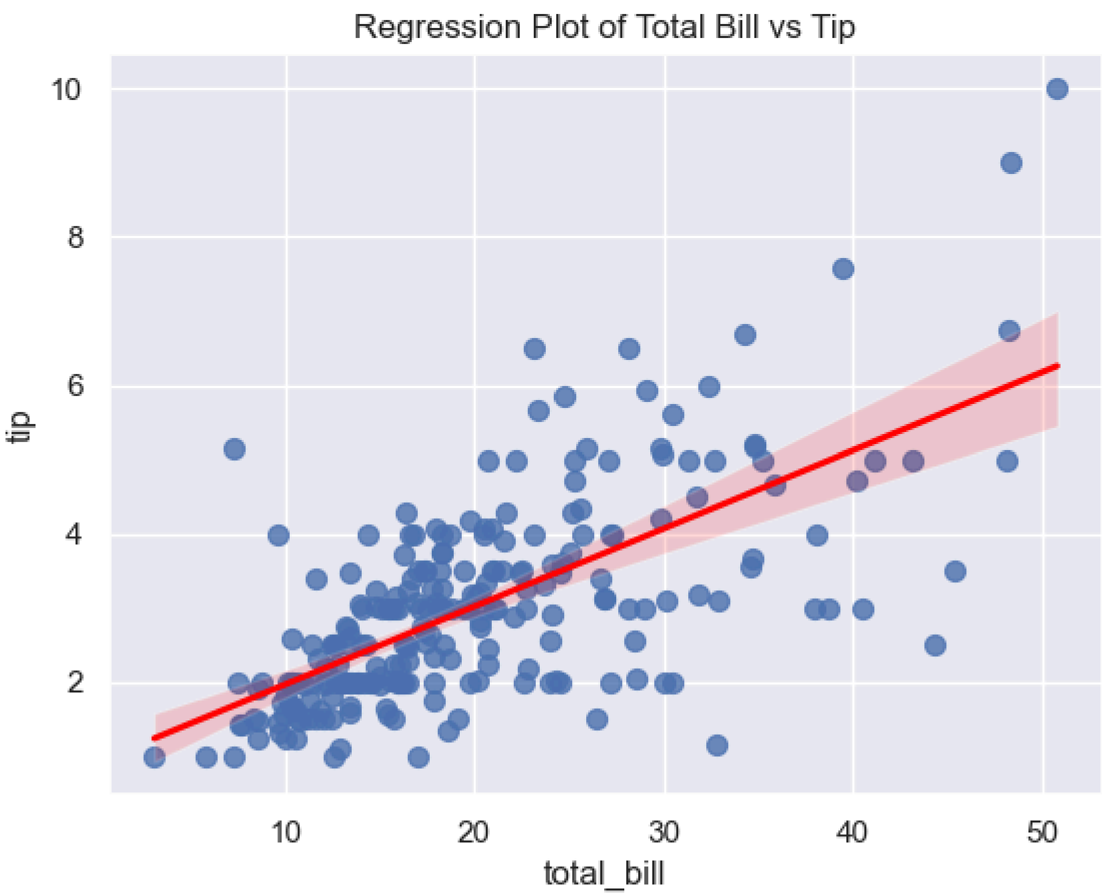
In [30]:
```
#6. count plot

sns.countplot(data=tips, x='day', hue='smoker', palette='dark')
plt.title("Count Plot of Days by Smoker Status")
plt.show()
```



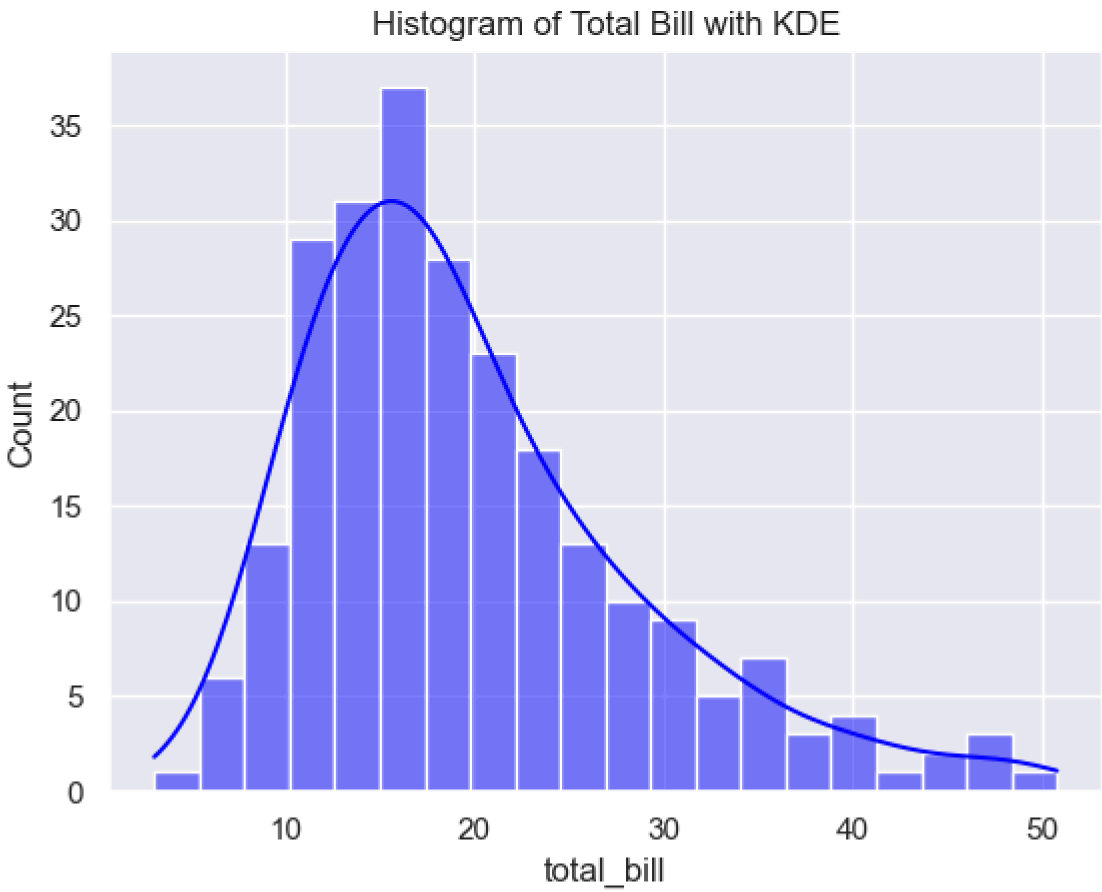
In [31]:
```
tips.columns
```

Out[31]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='obj
ect')
```

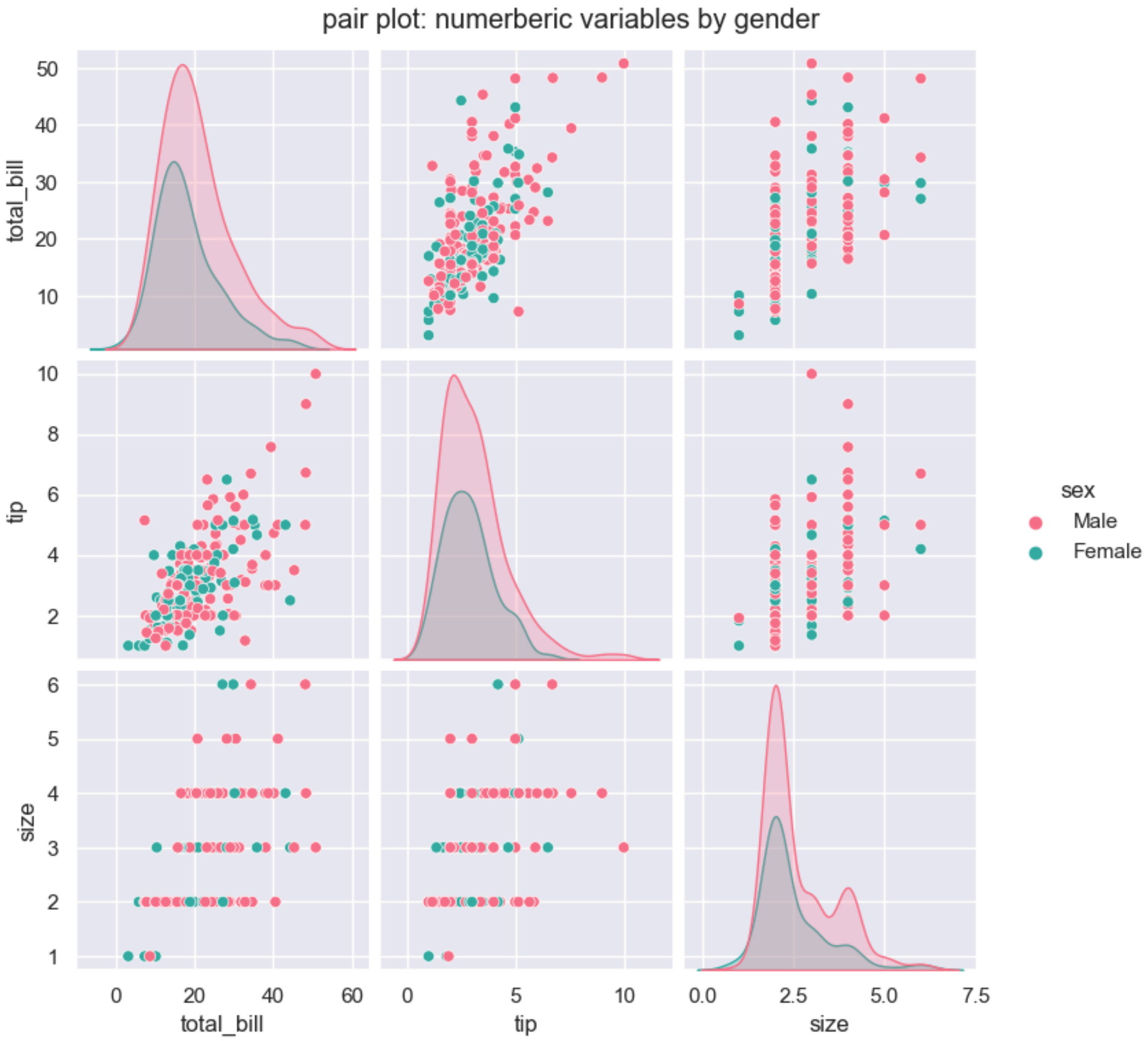
In [32]:
```
#7. regression plot

sns.regplot(data=tips, x='total_bill', y='tip', scatter_kws={'s':50}, line_kws={
plt.title("Regression Plot of Total Bill vs Tip")
plt.show()
```



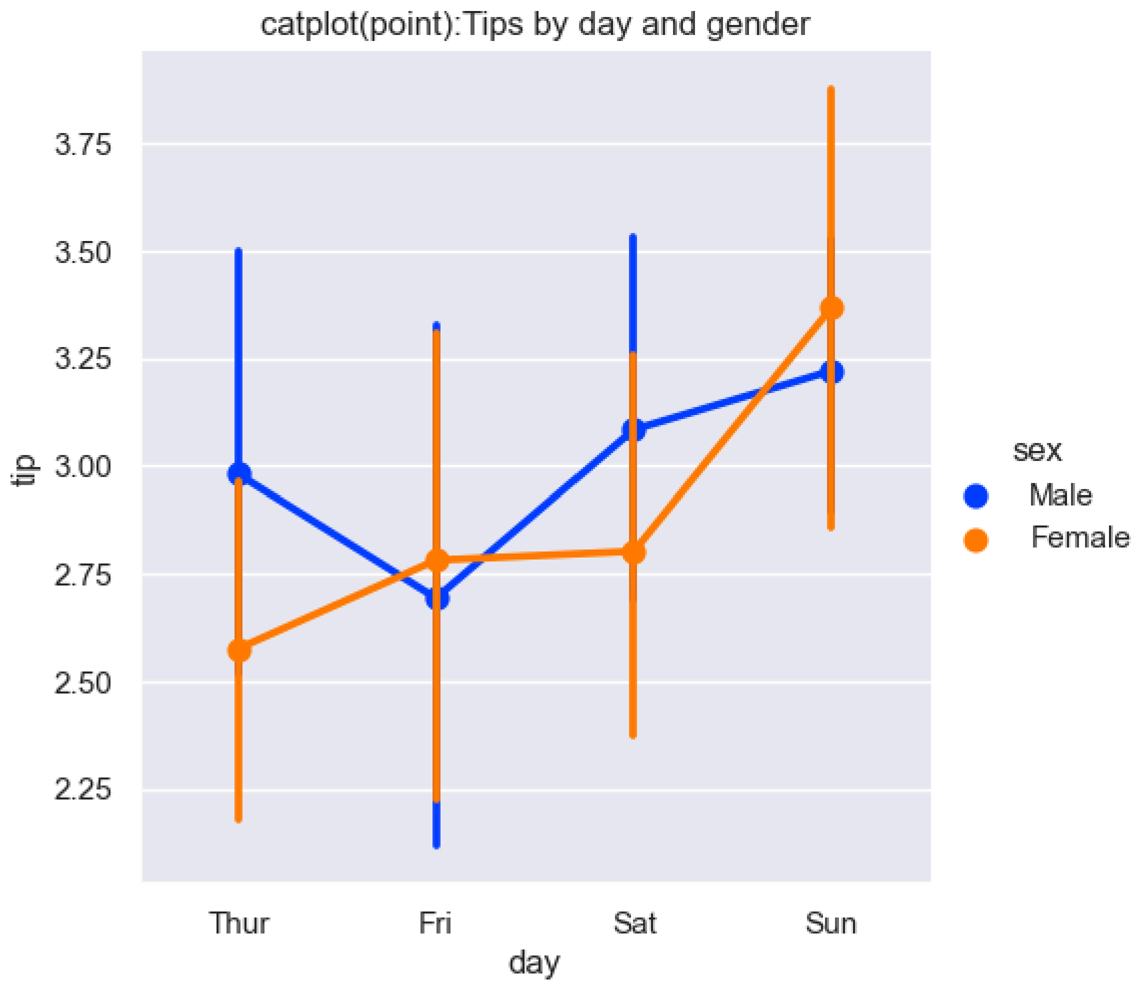
In [33]:
```
# 8. Histogram of total bill with KDE

sns.histplot(data=tips, x='total_bill', bins=20, kde=True, color='blue')
plt.title("Histogram of Total Bill with KDE")
plt.show()
```



In [34]:
```
#9. pairplot

sns.pairplot(tips, hue='sex', vars=["total_bill", "tip", "size"], palette='husl'
plt.suptitle("pair plot: numerberic variables by gender", y=1.02)
plt.show()
```

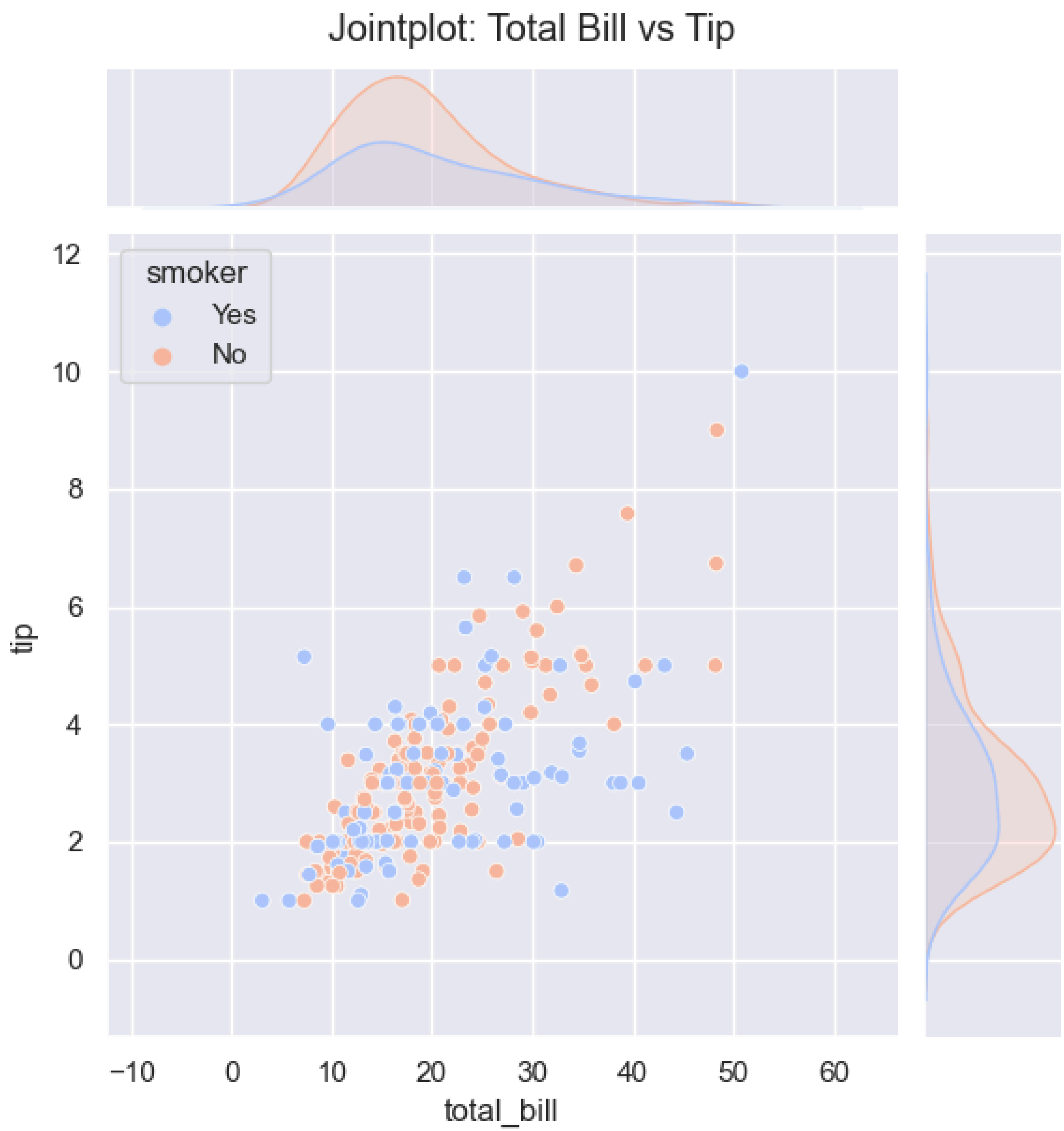


In [35]:
```python
# 10 catplot

sns.catplot(data=tips, x='day', y='tip', hue='sex', kind='point', palette='brigh
plt.title("catplot(point):Tips by day and gender")
plt.show()
```

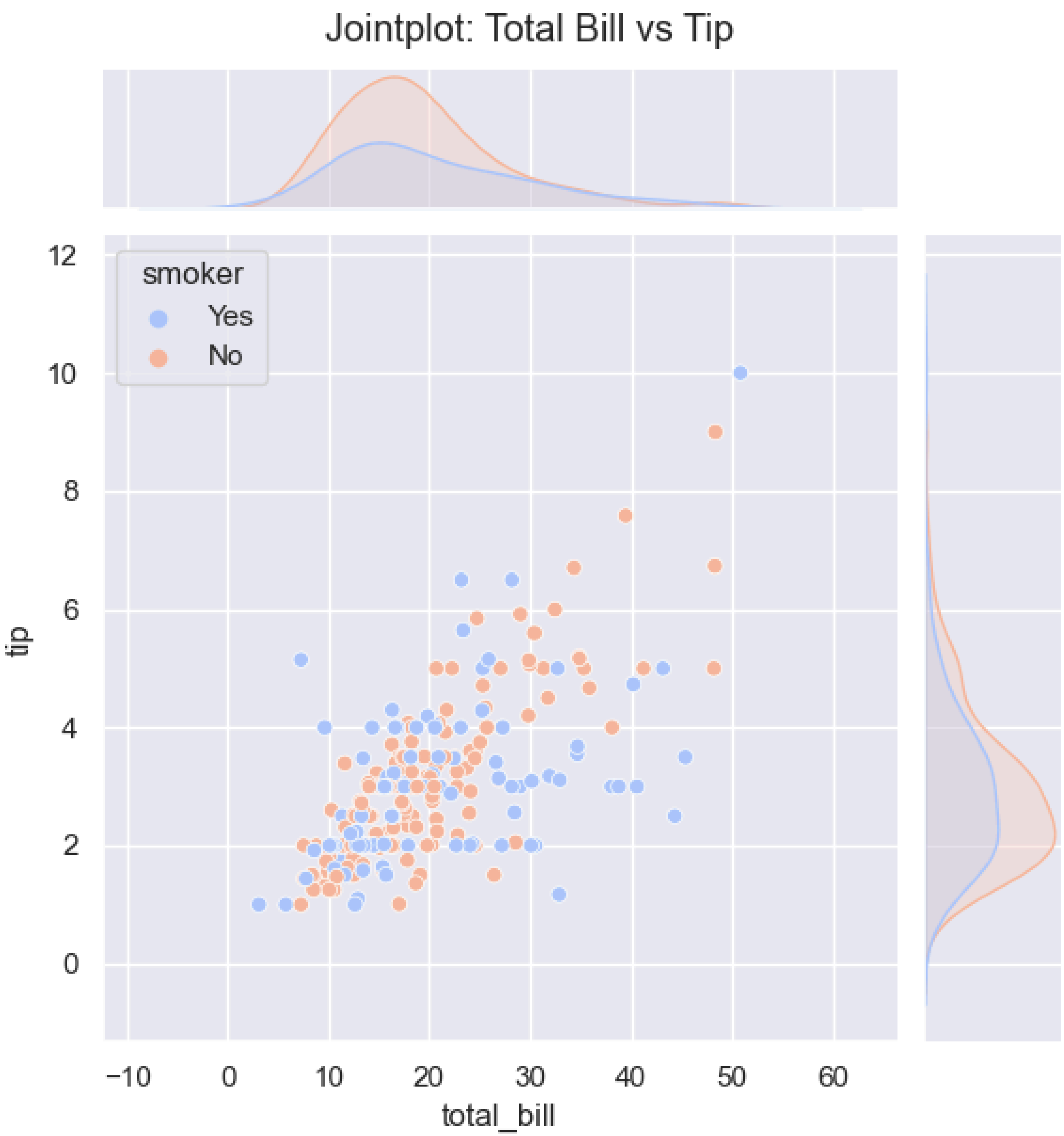


In [36]:

```
# 11. jointplot

sns.jointplot(data=tips, x='total_bill', y='tip', kind='scatter', hue='smoker',
plt.suptitle("Jointplot: Total Bill vs Tip", y=1.02)
plt.show()
```

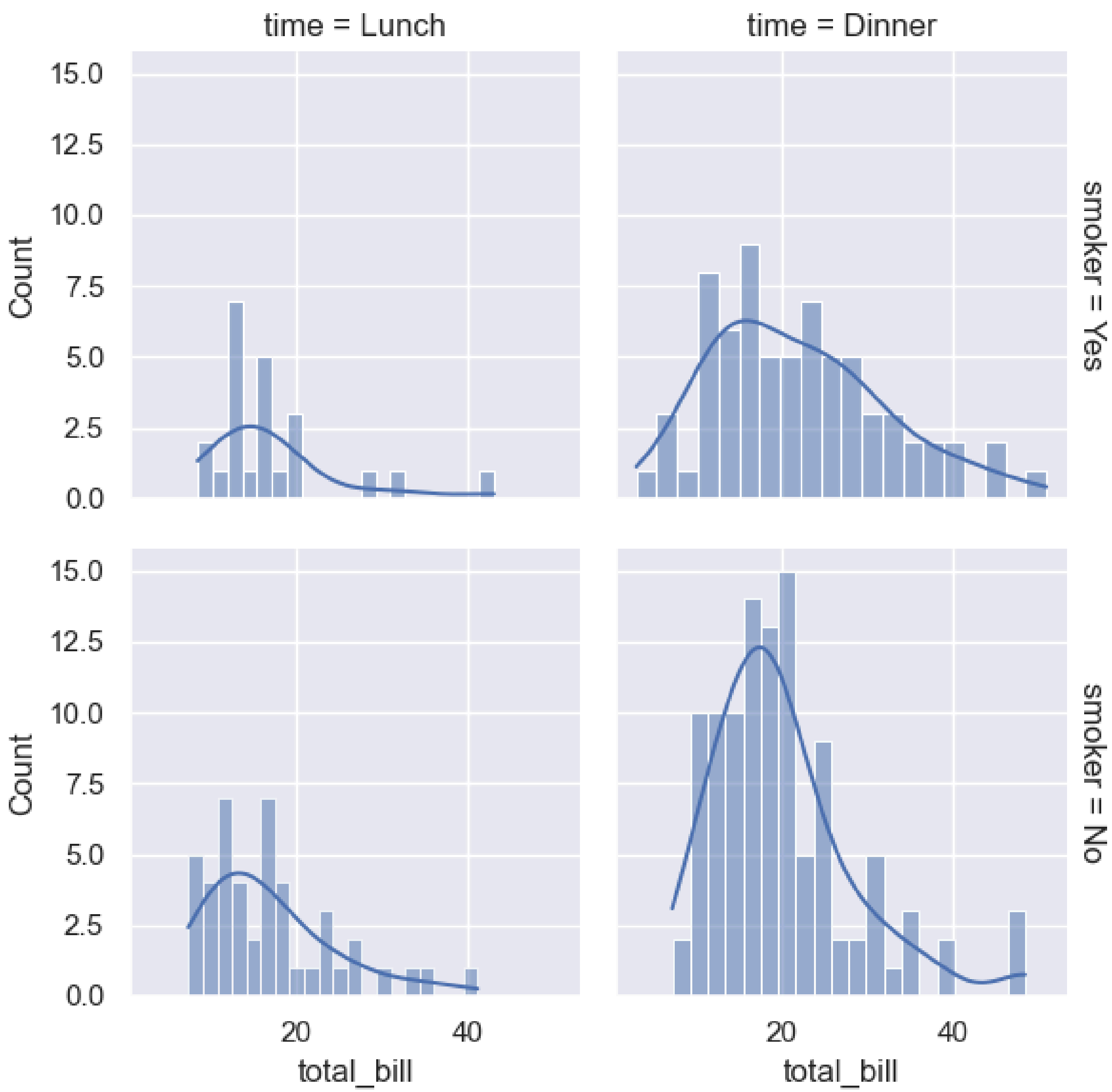


In [37]:
```
# 11. jointplot

sns.jointplot(data=tips, x='total_bill', y='tip', kind='scatter', hue='smoker',
plt.suptitle("Jointplot: Total Bill vs Tip", y=1.02)
plt.show()
```



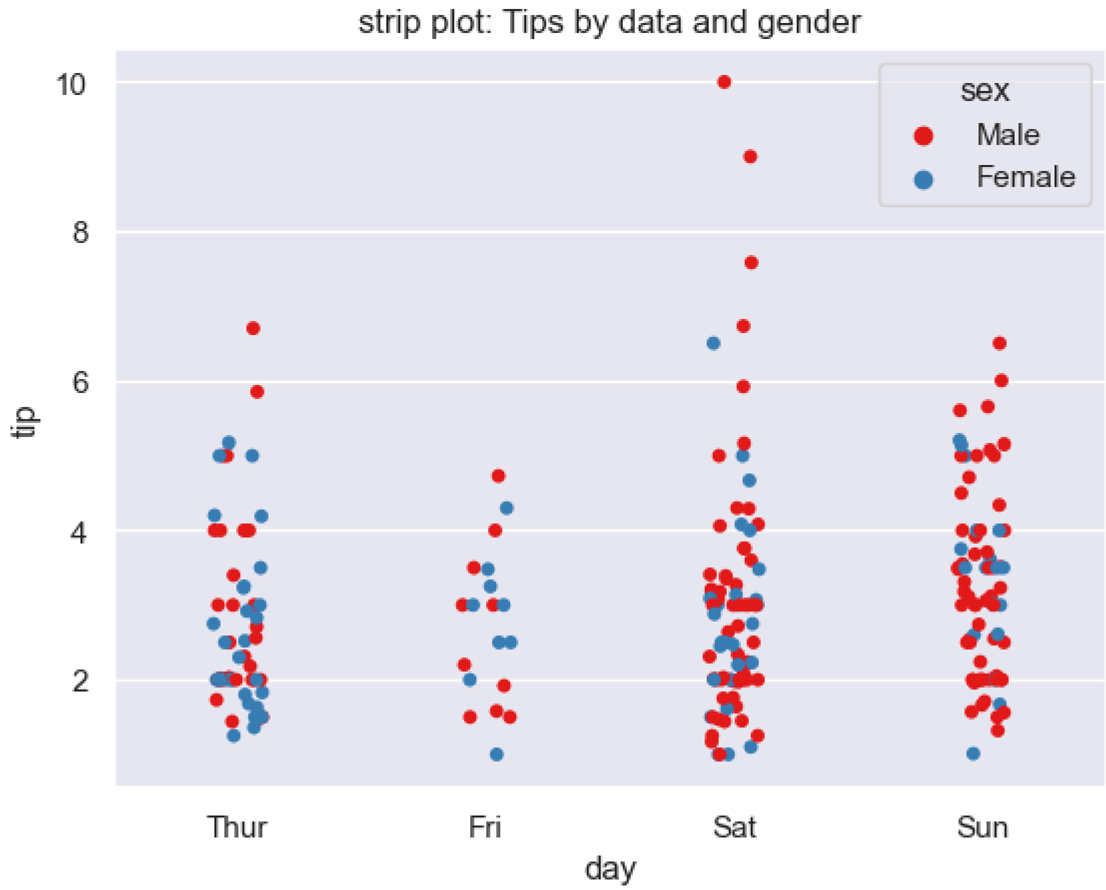
In [38]:
```
# Facetgrid

g = sns.FacetGrid(tips, col='time', row='smoker', margin_titles=True).map(sns.hi
g
```

Out[38]: <seaborn.axisgrid.FacetGrid at 0x233c087c450>



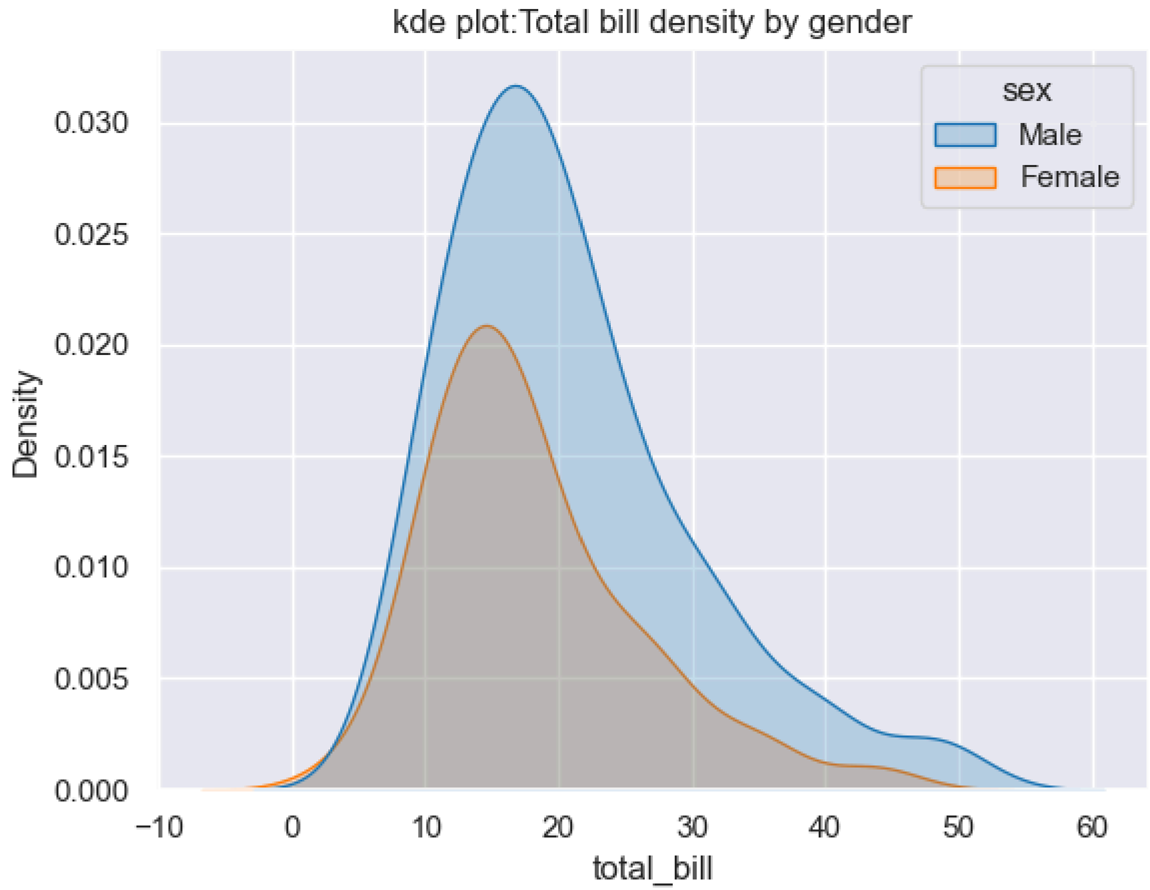
In [39]:

```
#13. strip plot

sns.stripplot(data=tips, x='day', y='tip', hue='sex', jitter=True, palette='Set1
plt.title("strip plot: Tips by data and gender")
plt.show()
```



In [40]:

```
# 14. KDE PLOT

sns.kdeplot(data=tips, x='total_bill',hue='sex', fill=True, palette='tab10')
plt.title("kde plot:Total bill density by gender")
plt.show()
```



In [41]:

```
import streamlit as st
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="whitegrid")

tips = sns.load_dataset("tips")

st.title('Mr.prakash senapati seaborn bootcamp tips data visualization app')
st.write("This is a simple app to visualize the tips dataset using seaborn.")

# function create and display plot
def display_plot(title, plot_func):
    st.subheader(title)
    fig, ax = plt.subplots(figsize=(8, 6))
    plot_func(ax=ax)
    st.pyplot(fig)
    plt.close(fig)

# plot

def scatter_plot(ax):
    sns.scatterplot(data=tips, x="total_bill", y="tip",hue="time", size="size",
    ax.set_title("Scatter plot of total bill vs tip")\

def line_plot(ax):
    sns.lineplot(data=tips, x= 'size', y='total_bill', hue='sex',markers='o',ax=
    ax.set_title("Line plot of total bill vs tip")

def bar_plot(ax):
    sns.barplot(data=tips, x='day', y='total_bill', hue = 'sex',palette='muted',
    ax.set_title("Barplot of Total Bill by Day")
```

```python
def box_plot(ax):
    sns.boxplot(data=tips, x='day', y='tip', hue='smoker', palette='Set2',ax=ax)
    ax.set_title("Boxplot of Tips by Day and Smoker Status")

def violin_plot(ax):
    sns.violinplot(data=tips, x='day', y='total_bill', hue='time', split=True, p
    ax.set_title("Violin Plot of Total Bill by Day and Time")

def count_plot(ax):
    sns.countplot(data=tips, x='day', hue='smoker', palette='dark',ax=ax)
    ax.set_title("Count Plot of Days by Smoker Status")

def reg_plot(ax):
    sns.regplot(data=tips, x='total_bill', y='tip', scatter_kws={'s':50}, line_k
    ax.set_title("Regression Plot of Total Bill vs Tip")

def hist_plot(ax):
    sns.histplot(data=tips, x='total_bill', bins=20, kde=True, color='blue',ax=a
    ax.set_title("Histogram of Total Bill with KDE")

def strip_plot(ax):
    sns.stripplot(data=tips, x='day', y='tip', hue='sex', jitter=True, palette='
    ax.set_title("strip plot: Tips by data and gender")

def kde_plot(ax):
    sns.kdeplot(data=tips, x='total_bill',hue='sex', fill=True, palette='tab10',
    ax.set_title("kde plot:Total bill density by gender")


display_plot("Scatter Plot", scatter_plot)
display_plot("Line Plot", line_plot)
display_plot("Bar Plot", bar_plot)
display_plot("Box Plot", box_plot)
display_plot("Violin Plot", violin_plot)
display_plot("Count Plot", count_plot)
display_plot("Regression Plot", reg_plot)
display_plot("Histogram Plot", hist_plot)
display_plot("Strip Plot", strip_plot)
display_plot("KDE Plot", kde_plot)

# Run from the console as steamlit run sns_app.py
```

```
2026-04-22 11:32:05.607
  Warning: to view this Streamlit app on a browser, run it with the following
  command:

    streamlit run C:\Users\Jagatjyoti\anaconda3\Lib\site-packages\ipykernel_launc
her.py [ARGUMENTS]
```